

Training a 30M-Parameter Small Language Model from Scratch for English Children’s Fables

A reproducible study: can a tiny model rival a large LLM on constrained story generation?

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Abstract

We train a **30M-parameter Llama-style language model from scratch** on the TF1-EN-3M children's-fable corpus and study, end to end and reproducibly, whether such a tiny model can approach the output quality of a 130x larger instruction-tuned LLM (Qwen3-4B) on a *constrained* generation task. Following modern scaling-law practice, we move from a deliberately under-trained baseline (loss 1.8) through a Chinchilla-aware retraining (loss 1.278, held-out perplexity 3.56) and a targeted data intervention that cures a template-collapse failure ("wise old owl" generation rate 90% -> 23%). We add an application-analysis dashboard (intrinsic diversity, readability, Zipf, positional loss) and close with a **systematic post-training campaign**: four alignment methods sharing one evaluation protocol - DPO (194 pairs), SFT-on-best, threshold-filtered RAFT (200 stories judged ≥ 9.0) and GRPO-lite (REINFORCE with a group baseline, 60 on-policy steps) - are **all null** on the model's default distribution, a fifth (knowledge distillation from the 4B teacher, 600 stories) *moves* the model but slightly degrades it (-0.37, the imitation trap), while **inference-time best-of-N search captures a confirmed +0.8 judge-point gain** (7.7 -> 8.55, near the 4B reference at 9.75) and ships in the app. Along the way we *measure* the LLM-judge's own noise (repeated evaluations of the same model differ by up to 0.45 at n=15) and show it retroactively explains two earlier false positives. The final 30M model generates coherent, complete fables at **~949 tokens/s, roughly 50x faster than the 4B reference**, reaching an LLM-judge quality of ~7.9/10 as measured under the final protocol (n=45). All parameters, curves and artifacts are reported from training step 0.

1. Introduction and thesis

Large language models dominate open-ended text generation, but their size makes them slow and costly. This project asks a narrower, defensible question: **on a well-scoped task (children's fables conditioned on five narrative slots), how close can a from-scratch Small Language Model (SLM) of ~30M parameters get to a 4B instruction-tuned LLM, and at what efficiency?**

The scientific priority of the project is the *training methodology*, not the app. Every claim is grounded in published method and evaluated with reproducible metrics. The narrative of the work is a sequence of hypotheses and measured outcomes:

1. A reduced baseline is intentionally under-trained (diagnosis).
2. Right-sizing the token budget per scaling laws fixes it (Phase 1).
3. A data intervention fixes a specific qualitative failure (Phase 2).
4. Sampling temperature cannot fix prompt-adherence (measured); we hypothesized preference optimization could.
5. A four-method post-training campaign (DPO, SFT-on-best, RAFT, GRPO-lite) tests that hypothesis rigorously - and refutes it at this scale and feedback budget, while inference-time best-of-N search delivers the gain the training methods could not.

2. Scientific grounding

- **Autoregressive LM = MLE on the chain rule** $\log p(x) = \sum_i \log p(x_i | x_{<i})$ (course Week 6). We loss-mask the conditioning so the model only maximizes the log-likelihood of the *story* tokens.
- **Scaling laws (Kaplan et al. 2020)**: test loss falls as a power law in parameters N , tokens processed D , and compute C . Under-training = too small a D for a given N .
- **Chinchilla (Hoffmann et al. 2022)**: compute-optimal ~20 tokens per parameter.
- **Data-constrained scaling (Muennighoff et al. 2023)**: repeating data up to ~4 epochs is nearly as good as fresh data.
- **DPO (Rafailov et al. 2023) / RLAI (Bai et al. 2022)**: align a model to preferences using pairwise (chosen, rejected) data, here labeled by an AI judge.
- **REINFORCE + baseline (Williams 1992; course Week 10)**: policy-gradient RL with variance reduction by subtracting a baseline from the reward; normalizing within a group of rollouts per prompt yields the GRPO estimator (Shao et al. 2024) used by DeepSeek-R1. This grounds our GRPO-lite experiment (Section 9.12).
- **RAFT (Dong et al. 2023)**: keep only high-absolute-reward samples and fine-tune on them - the design of our threshold-9.0 experiment (Section 9.10).
- **Dataset + evaluation methodology**: TF1-EN-3M (Nadas et al. 2025, arXiv:2504.20605); evaluation follows our ADR-0002 (objective metrics + a cross-family LLM-judge panel).

3. Dataset

We use `klusai/ds-tf1-en-3m`. Each record has a prompt (five scaffold slots: Main Character, Setting, Challenge, Outcome, Teaching/Moral) and a fable. We format each training example as a **conditional sequence**:

```
<conditioning: the 5 slots + a length hint>
<|story|> <fable text> <|end|>
```

Only the story region contributes to the loss (the conditioning prefix is masked with -100). Key data-pipeline choices:

- **Quality filter:** keep fables of 60-320 words (avoid truncated / rambling extremes).
- **Slot dropout:** during training each slot is randomly blanked so the model learns to generate with any subset of the five slots present.
- **Custom BPE tokenizer, vocab 12k**, trained on the fable corpus - keeps the embedding table small, which matters at 30M parameters.

4. Model and training recipe

The architecture is a standard Llama-style decoder (RoPE, Grouped-Query Attention, RMSNorm, SwiGLU, tied input/output embeddings).

Component	Value
Parameters	~36.6M
Hidden size / FFN	512 / 2048
Layers / heads / KV heads	8 / 8 / 2 (GQA)
Vocab / sequence length	12,000 / 512
Optimizer	AdamW, betas (0.9, 0.95), weight decay 0.1
Grad clip	1.0
LR schedule	Warmup-Stable-Decay (warmup 2%, decay 20%)
Peak LR	3e-3
Effective batch	32 x 4 accum = 128 sequences (~33k story tokens/step)
Precision	fp16 (T4 GPU)

Architectural ceiling. The model is trained at sequence length 512. With a conditioning prompt of ~50-110 tokens this leaves ~400-460 tokens for the story, which caps the maximum coherent fable length. This ceiling later drives the “story completeness” design (Section 9.6).

5. Experiments (from step 0)

Run	Data	Steps	Final loss	Held-out PPL	Judge (overall)	Notes
v1 (baseline)	150k	900	~1.80	-	2.5	intentionally under-trained (~1.7 tok/param)
Phase 1	400k v1	1800	1.447	4.18	6.0	right-sized token budget
+ sampling fix	-	-	-	-	6.2	repeat_penalty 1.3 -> 1.1 (entity drift)

Run	Data	Steps	Final loss	Held-out PPL	Judge (overall)	Notes
Phase 2	400k v2	3600	1.278	3.56	7.0	data intervention (resume 1800 -> 3600)
Phase 2 + DPO	194 pairs	30	(loss 1.278)	3.54	7.88 (n=15)	null vs baseline 8.02 (Section 9.8)
+ RAFT	200 stories >= 9.0	30	0.672 (SFT)	10.64*	7.60 (n=15, pooled)	null (Section 9.10)
+ GRPO-lite	60 steps x 16 rollouts	30	on-policy	10.58*	8.03 (n=45)	null, KL ~1e-3 (Section 9.12)
Best-of-3 (no training)	-	30	-	-	8.55 (n=15)	deployed in app (Section 9.9)
Qwen3-4B (ref)	-	-	-	-	9.75	130x larger, instruction-tuned

*Perplexity for the alignment rows is measured on a different held-out slice (raw `test.jsonl` text) than the 3.5x figures above; only the *relative* drift matters.

Phase 2 uses two data interventions motivated by qualitative analysis (Section 9.4): cap the “wise old owl” template to 10% of the corpus, and lower the slot dropout for the Teaching/Outcome slots (0.30 -> 0.15) so the model learns to follow the requested moral.

6. Training dynamics

The loss falls smoothly from ~7 to **1.278**. The log-log fit gives an exponent of about -0.25 with $R^2 \sim 0.96$, empirical evidence that our from-scratch run follows the scaling-law power law rather than diverging.

7. Language-modeling quality

Held-out perplexity of **3.56** is within 1% of the floor $e^{(1.278)} = 3.59$, meaning the model’s held-out behavior matches its training loss - no over-fitting despite the small size, thanks to the large unique corpus.



Figure 1: Training loss over the full run. Phase 2 resumes at step 1800 on the cleaned corpus v2; the WSD decay pulls the final loss into the target band (< 1.5).

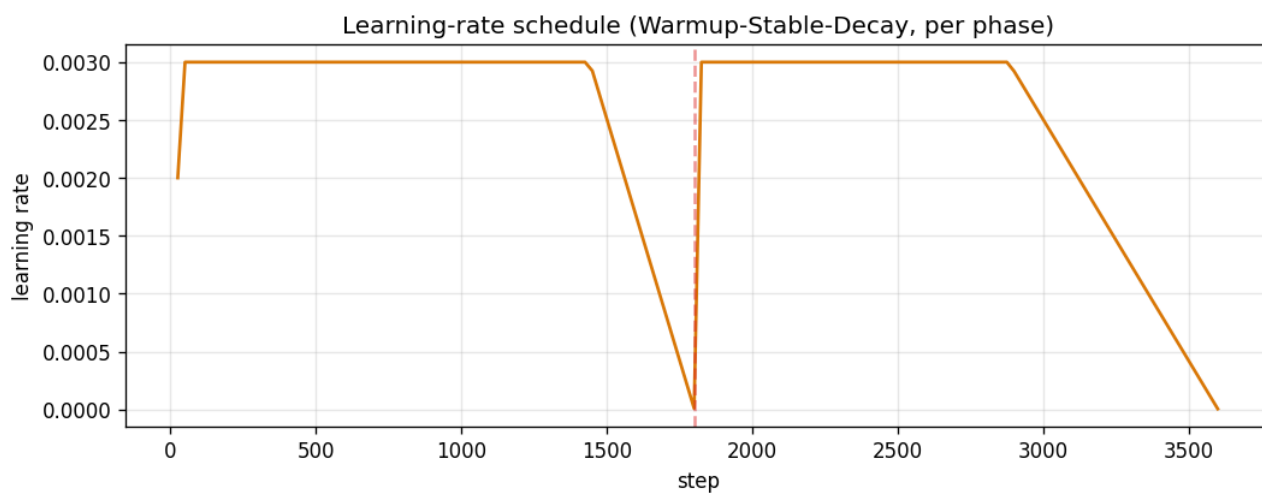


Figure 2: WSD learning-rate schedule (per phase).

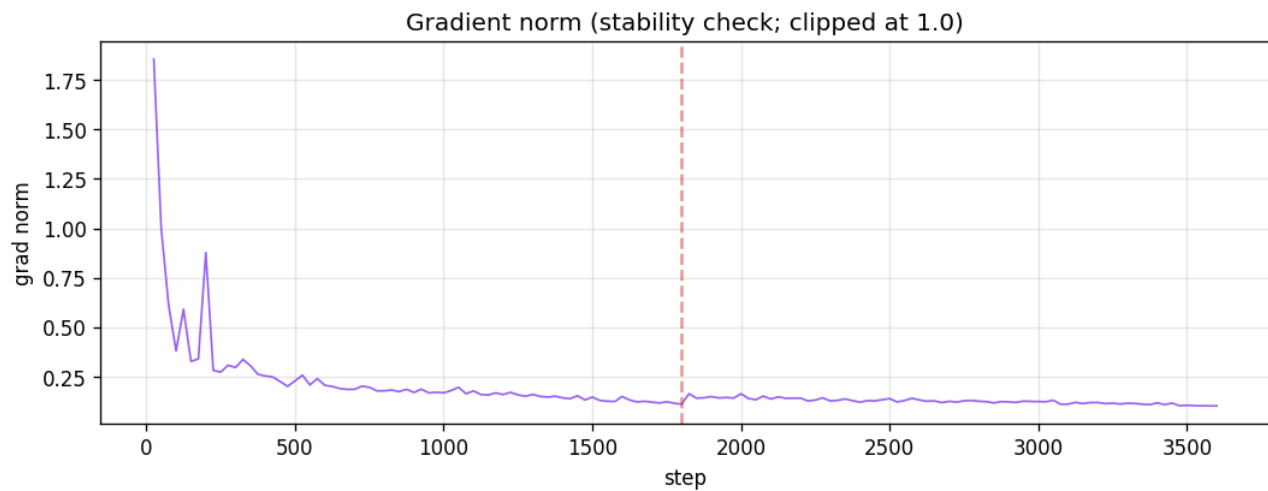


Figure 3: Gradient norm stays controlled (clip 1.0) - no instability across either phase.

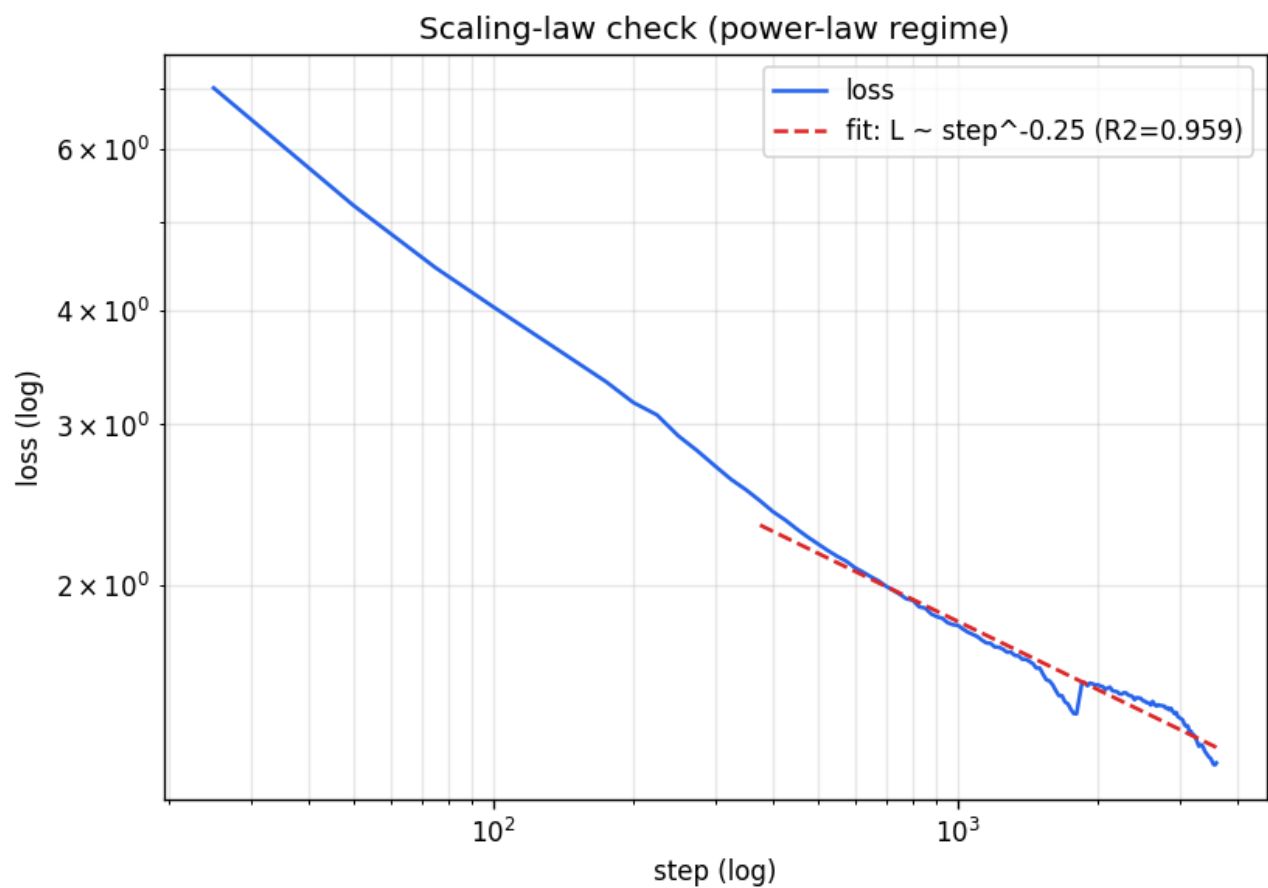


Figure 4: Scaling-law check: on log-log axes the post-warmup loss is near-linear ($R^2 \sim 0.96$), i.e. the run stays in the power-law regime predicted by Kaplan et al. (2020).

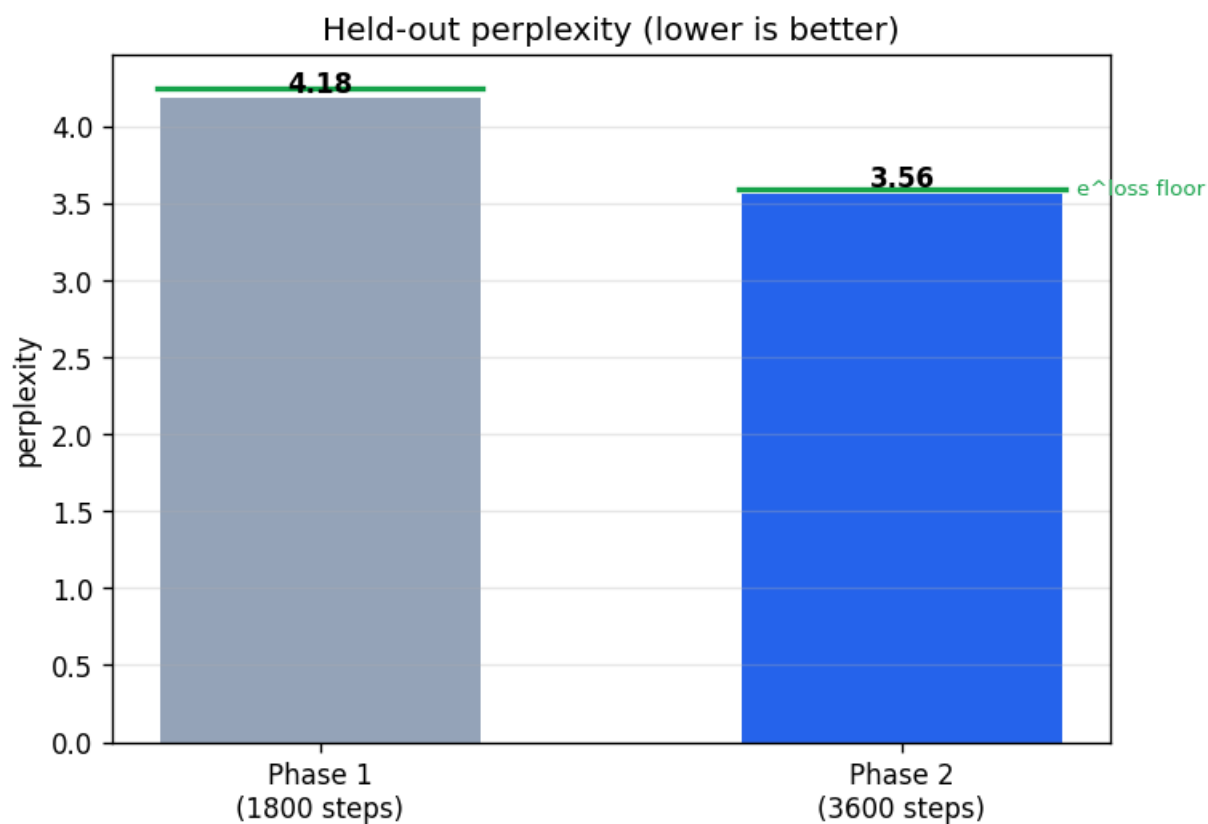


Figure 5: Held-out perplexity. Phase 2 (3.56) improves on Phase 1 (4.18); both sit essentially on the theoretical floor $e^{\text{train loss}}$, so the model generalizes without over-fitting.

8. Application analysis (intrinsic, reference-free)

Following ADR-0002, we compute reference-free metrics on generated stories and compare against real held-out fables.

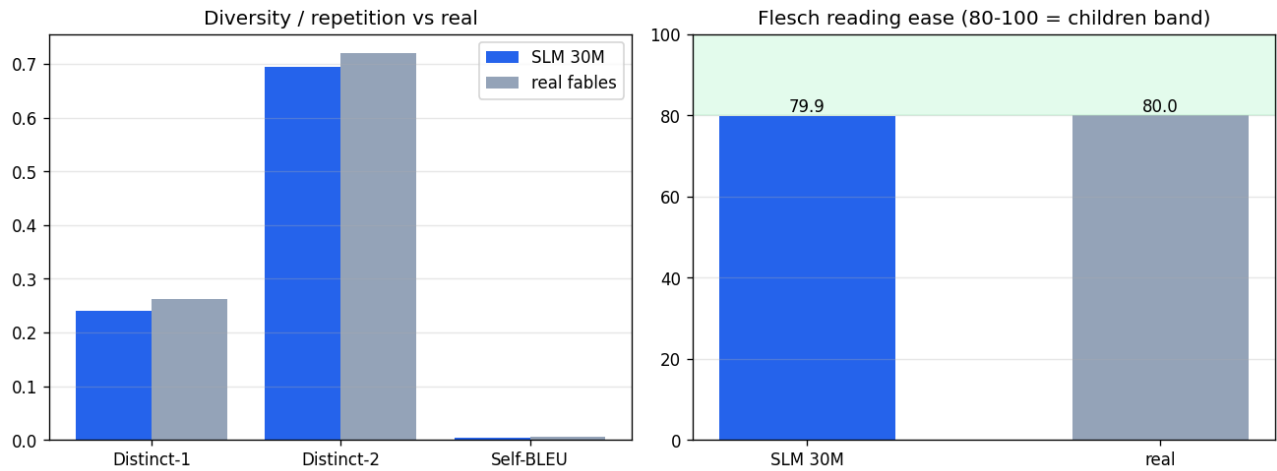


Figure 6: Diversity (Distinct-1/2), repetition (Self-BLEU) and readability (Flesch) of generated vs real fables. Generated diversity sits within a few percent of real; Flesch matches the real corpus (~80).

The generated text matches real fables on diversity (Distinct-2 gap ~4%), repetition (Self-BLEU gap ~0.001), readability (Flesch 79.9 vs 80.0) and the Zipfian vocabulary profile - the model has learned the *statistical shape* of the domain, not just surface fluency.

9. Findings by stage

9.4 Data intervention: curing template collapse

Qualitative review of Phase 1 revealed **mode amplification**: the phrase “wise old owl” appears in 28% of *real* fables but the model emitted it in ~90% of generations. Capping that template to 10% of the corpus (Phase 2) reduced the generation rate to **23%**, below even the data prior.

9.5 Quality progression

9.6 Story completeness (a limitation turned into a fix)

The Phase-2 model occasionally cut stories mid-sentence at short lengths. Diagnosis: the generation cap (300 tokens) was smaller than the model’s natural fable length, so it hit the cap before emitting `<|end|>`. Because the model is trained at sequence length 512, simply “adding tokens” is not an option beyond ~460. The fix: right-size the per-length token budget to the architectural ceiling, use `done_reason` to detect true completion vs cut-off, and trim any residual cut to the last complete sentence. Result: **30/30 test stories complete** with a real ending.

9.7 Prompt-adherence: sampling vs alignment

We measured whether sampling temperature affects slot adherence. On 10 held-out prompts, temperature 0.7 and 0.8 gave *identical* adherence (69%): **sampling is not the lever**; adherence is

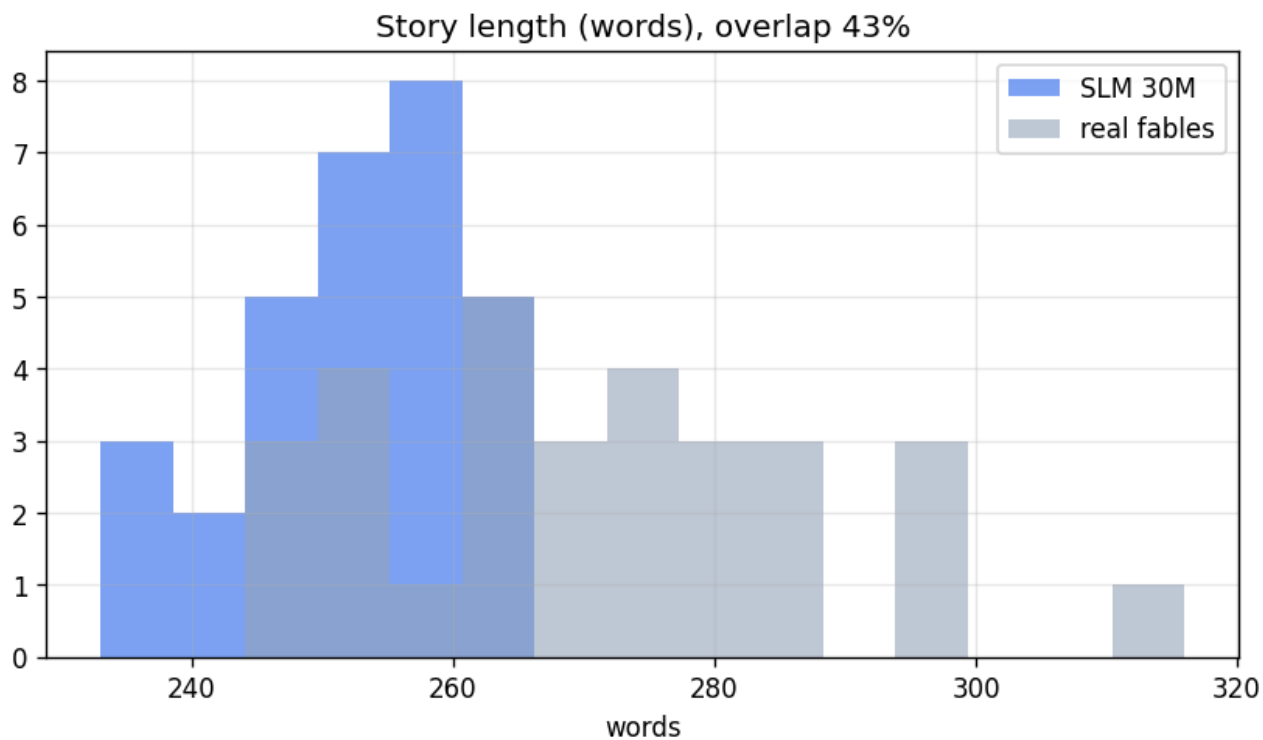


Figure 7: Generated story-length distribution vs real fables.



Figure 8: Mean cross-entropy by relative position in the story - the model is most confident at the opening and stays stable through the body.

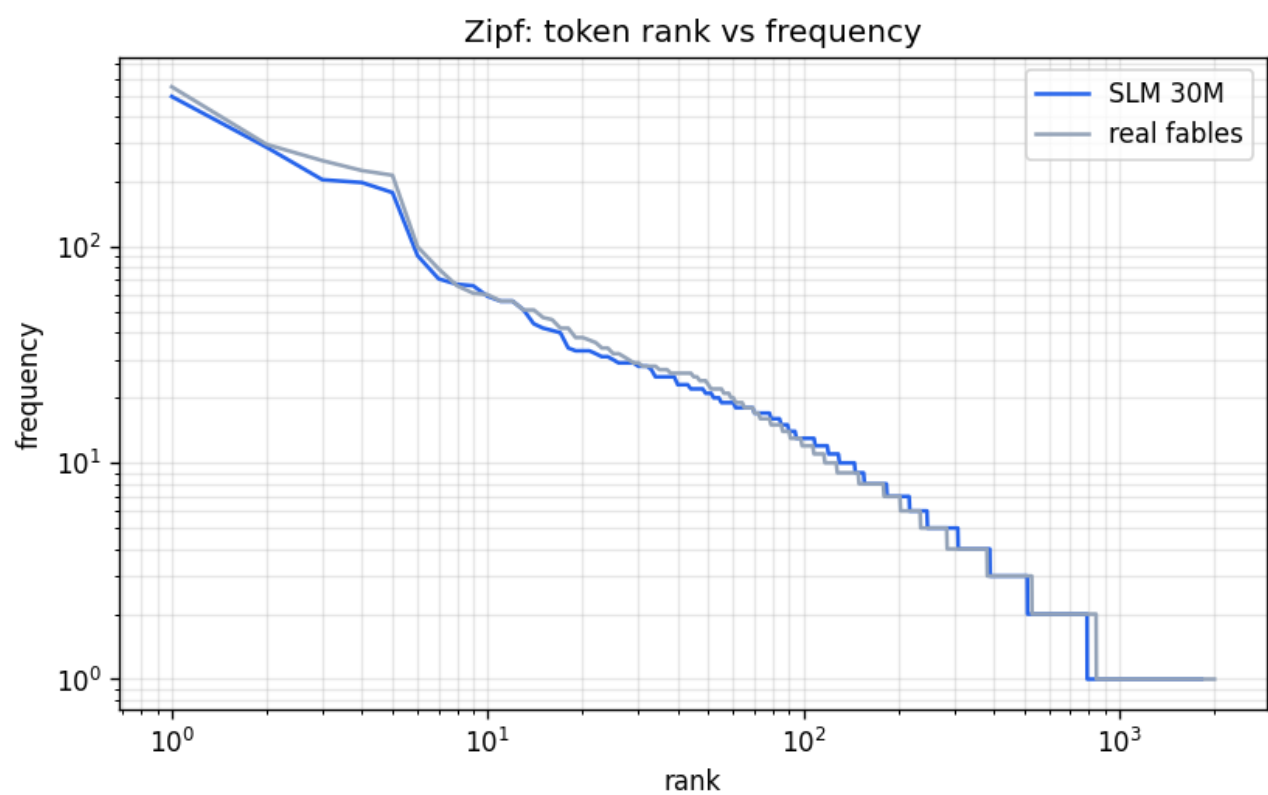


Figure 9: Zipf rank-frequency of generated tokens closely tracks the real-fable vocabulary distribution.

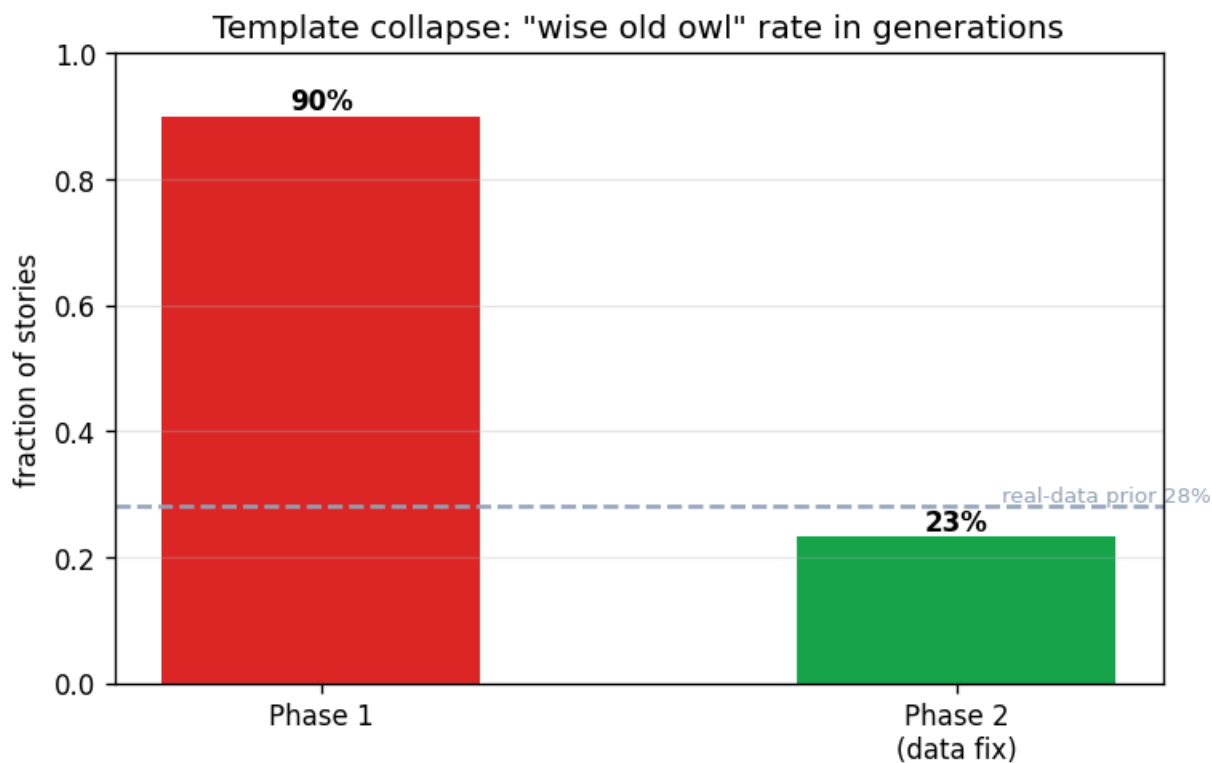


Figure 10: Template collapse before/after the data intervention.

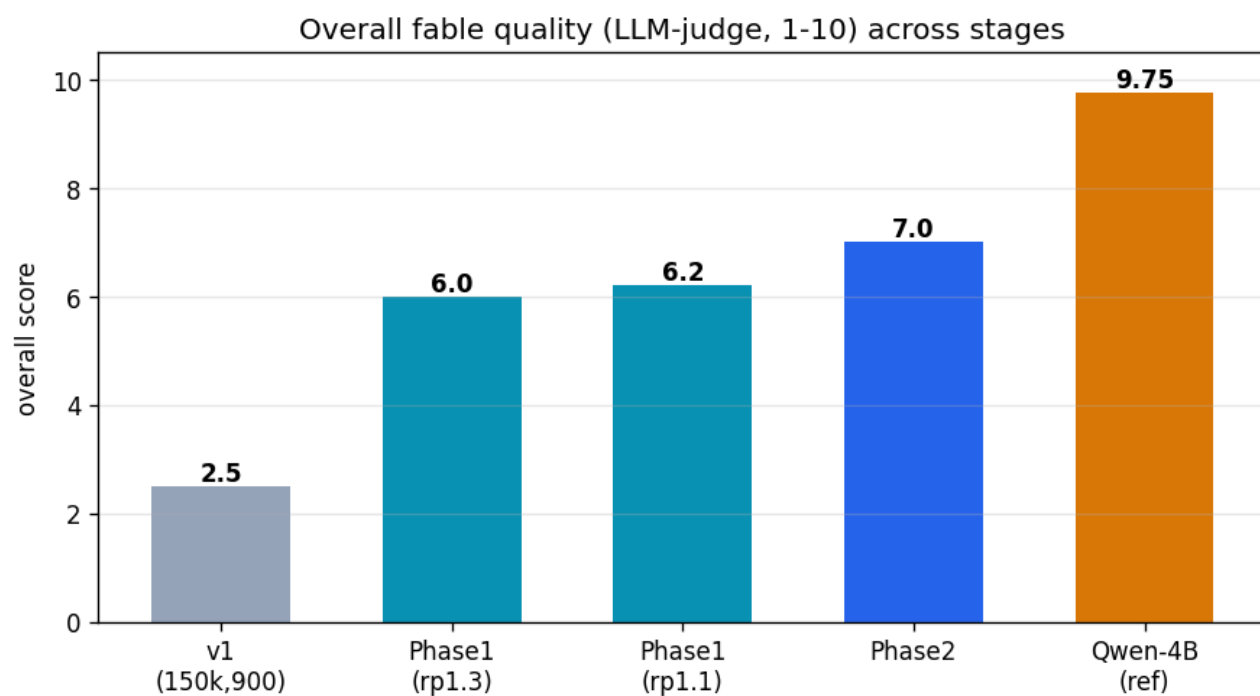


Figure 11: Overall LLM-judge quality across stages, with the 4B reference for scale. The 30M model rises from 2.5 (under-trained) to ~7.0, closing much of the gap to Qwen-4B (9.75) at 1/130th the size.

capped by the 30M model’s weak conditioning. At the time we hypothesized preference optimization was the proper fix - the campaign below tests that hypothesis.

9.8 DPO preference alignment (RLAIF): an instructive null

We generate two stories per prompt from the Phase-2 model, have an AI judge score them on the four axes, and keep pairs with a clear preference (margin ≥ 1.0) as (chosen, rejected) data - 194 pairs after filtering. We then run **DPO** (ORPO was unavailable in the installed TRL 1.8; DPO is the canonical equivalent) locally on Apple-Silicon MPS.

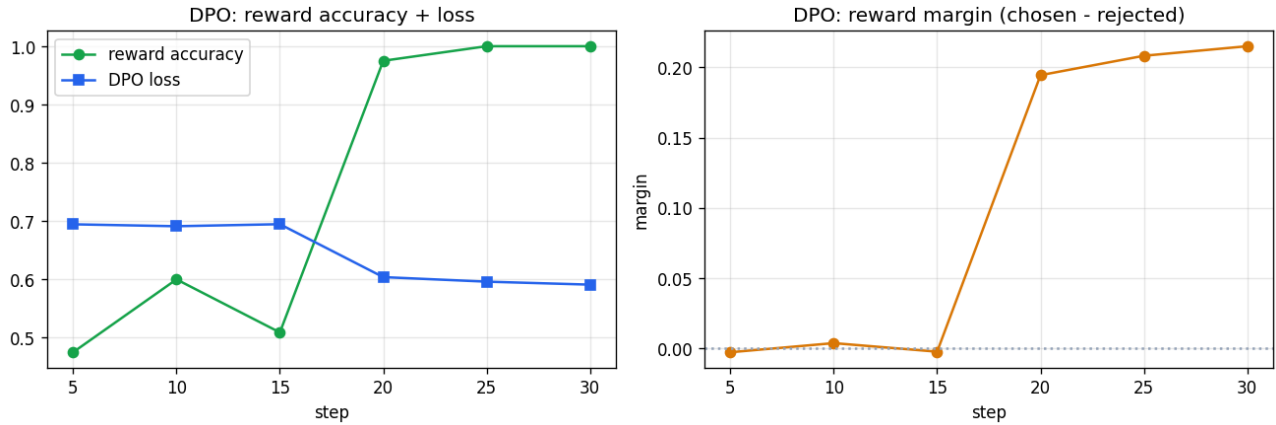


Figure 12: DPO reward accuracy climbs to 1.0 and the reward margin (chosen - rejected) turns strongly positive - the model learns the preference *on its training pairs*. Held-out perplexity is unchanged (0% drift): no catastrophic forgetting.

The in-training signal looks perfect - reward accuracy 1.0, positive margin, zero perplexity drift. An early slot-recall probe even suggested +5 points of adherence (71% $\rightarrow$ 76%, Fig. below). **Both impressions failed the rigorous test.** Under the standard protocol (15 held-out prompts, LLM-judge, fixed seeds), the DPO model scores **7.88 vs the baseline’s 8.02** - a null result. Section 9.11 shows the +5-point probe was within measurement noise. The mechanism-level explanation: with chosen and rejected drawn from the *same* model at similar quality, the relative preference signal is too weak to move the default distribution.

9.9 The headroom probe: best-of-N shows capacity is not the ceiling

Is the 30M model *unable* to produce great fables, or merely *inconsistent*? We sample K=3 candidates per prompt (temperatures 0.5/0.8/1.1) and let the judge pick the best:

Measure	Value
Single-sample mean (15 held-out prompts)	7.72
Best-of-3 mean	8.55
Individual best samples	up to 9.0-9.5 (4B reference: 9.75)

The model already *contains* near-reference-quality fables - the binding constraint is **variance, not capacity**. This result reframed the whole alignment effort: the goal is not to teach the model something new but to shift probability mass toward its own best modes. Best-of-N (off / 3 / 5) ships in the

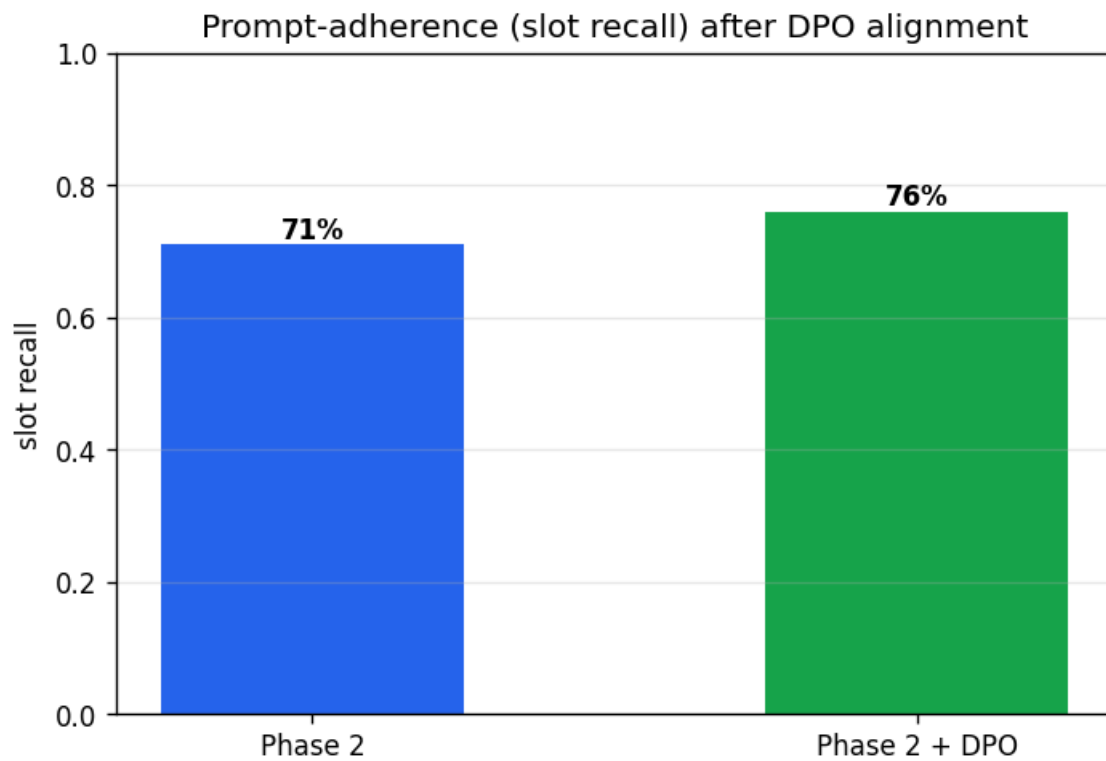


Figure 13: The early slot-recall probe (71% -> 76%) that later proved to be within judge noise - kept here as a documented lesson in evaluation rigor.

app as a user-facing control: the backend generates N candidates, the judge scores each, and the best is returned with all candidate scores logged in the Activity Log.

9.10 RAFT: threshold-filtered SFT at 5x scale - null

If best-of-N finds the good samples, can we *train on them* and internalize the gain? Our first SFT-on-best trial (42 stories) was null, but 93% of that corpus already scored ≥ 8.5 , so the untested variables were **scale** and a strict **absolute threshold** - the defining features of RAFT (reward-ranked fine-tuning). We built a 200-story corpus in which *every* story is judged ≥ 9.0 (mean 9.22; 105 harvested from earlier experiments, 95 newly generated with a 23% prompt acceptance rate), and fine-tuned at lr 2e-5 for 3 epochs with the conditioning loss-masked.

Result: **null again** - 7.60 (pooled over two evaluation runs) vs baseline 7.78, with perplexity drift +0.5% only. The theoretical reading: best-of-N samples are drawn from the model's *own* distribution, so supervised fine-tuning on them mostly re-weights modes the model already prefers; ~60k story tokens against a 600M-token pretraining prior is also a very small nudge. Crucially, the gradient contains no *negative* component - nothing pushes probability away from mediocre modes.

9.11 Measuring the judge itself: the noise that manufactured two false positives

Re-evaluating the *same* RAFT model twice (same protocol, same seeds) returned 7.38 and 7.82 - a 0.44 spread. The same double-measurement on the baseline returned 7.73/7.82, and on the GRPO model 8.00/8.45. At n=15 prompts, **the judge's own noise is $\sim \pm 0.4$** , which retroactively explains both the "DPO +5 adherence" probe (Section 9.8) and an early "GRPO +0.45" reading (Section 9.12) as sampling artifacts.

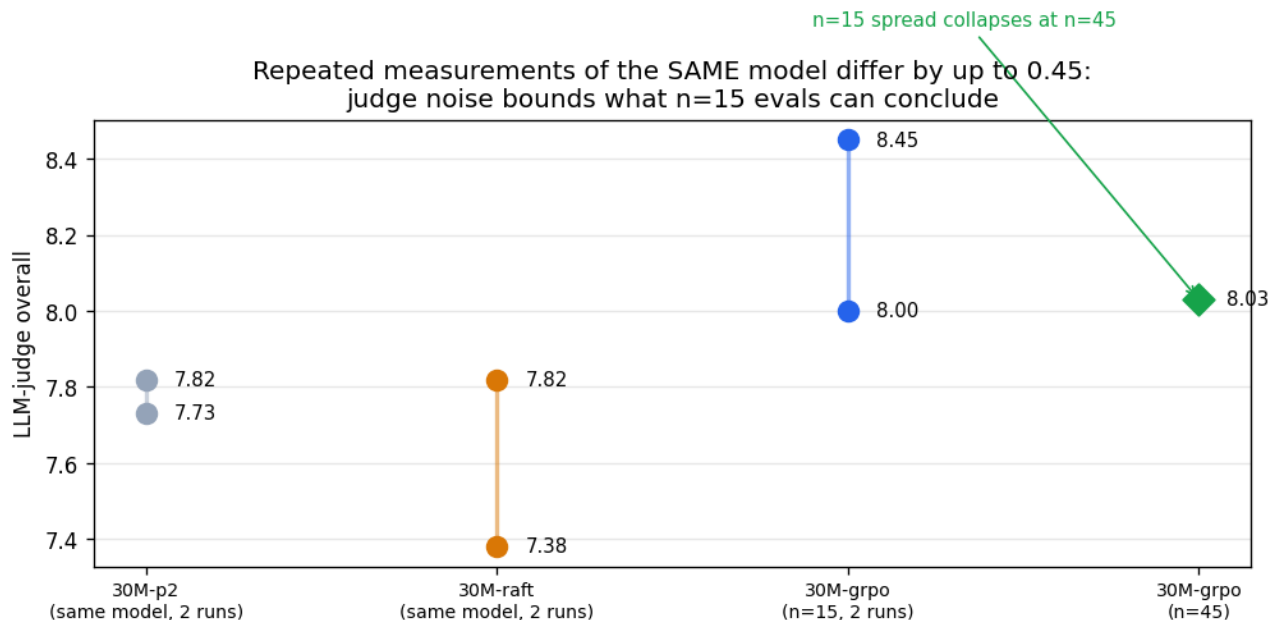


Figure 14: Repeated evaluations of the same checkpoints: the spread within a model is as large as the effects we were trying to detect at n=15.

Methodological rule adopted for all conclusions in this report: any delta below ~ 0.5 judge-points at n=15 is treated as noise; decisive comparisons are re-run at n=45 with paired seeds.

9.12 GRPO-lite: on-policy RL with a group baseline - null at this budget

The last untested ingredient class from the course material (Week 10: policy gradients, REINFORCE, variance reduction with a baseline) is **on-policy RL with an absolute reward and a negative gradient**. We implemented GRPO-lite: each step samples 4 prompts x 4 fresh rollouts, scores each rollout with the judge, normalizes the advantage within the group ($(r - \text{mean}) / \text{std}$ - the Week-10 baseline trick), and applies REINFORCE with a KL penalty (beta 0.05) against the frozen Phase-2 reference. Sixty steps (~960 judge calls, ~5 h, lr 3e-6 then 1e-5), checkpoint-resumed across interruptions.

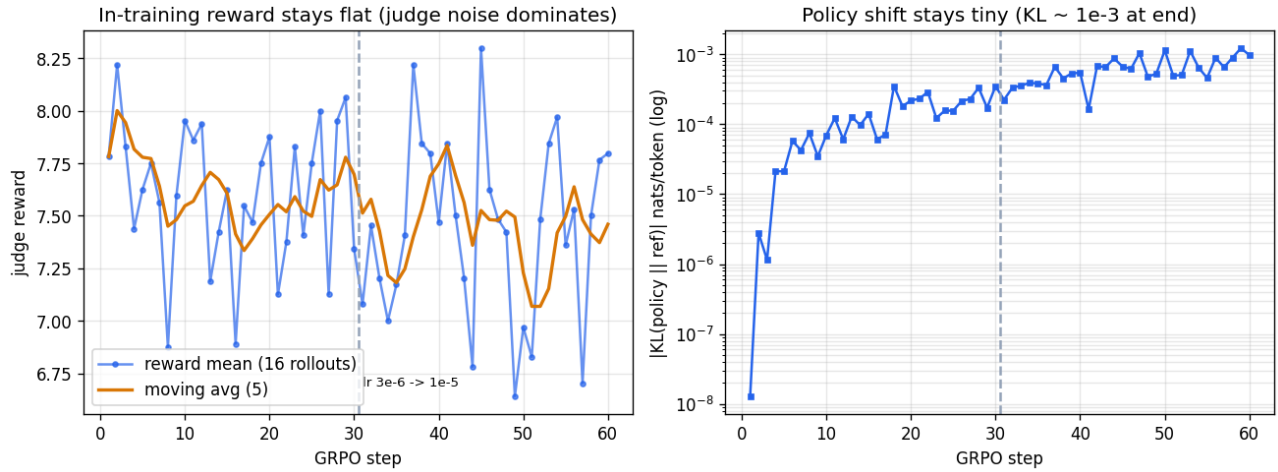


Figure 15: GRPO training dynamics: the in-training reward is dominated by judge noise, and the policy’s KL divergence from the reference stays around 1e-3 nats/token - the policy barely moves.

At $n=15$ the GRPO model read +0.45 above baseline - promising. Applying our own noise rule, we extended the evaluation to **$n=45$ paired prompts**: the delta collapsed to **+0.09 ($t=0.54$; win/tie/loss 17/10/18)**. Null - but a *diagnosable* one: with final KL $\sim 1e-3$, the policy never moved far enough for any effect to be measurable. The honest conclusion is “GRPO at a ~960-judge-call budget does not shift the distribution”, not “GRPO cannot”. The practical blocker is reward cost: at ~15 s per judge call, DeepSeek-R1-scale step counts are out of reach locally, and a distilled 30M reward model failed its validation gate (pairwise accuracy 46.7% ~ chance) - the quality signal is not learnable from ~500 noisy labels at this scale.

9.13 The campaign in one picture

Method	Signal	Exploration	Negative gradient	Result
DPO (194 pairs)	relative preference	no (static data)	implicit	null (7.88 vs 8.02)
SFT-on-best (42)	best-of-batch	no	no	null (7.98 vs 8.02)
RAFT (200 $\geq$ 9.0)	absolute threshold	no	no	null (7.60 vs 7.78)
GRPO-lite (60 steps)	group advantage	yes (on-policy)	yes	null at this budget (+0.09, $n=45$)
Distill-from-teacher (600)	off-distribution imitation	teacher’s	no	negative (-0.37, $n=45$; ppl +4.4%)

Method	Signal	Exploration	Negative gradient	Result
Best-of-N (inference)	judge selection	yes (test-time)	-	+0.8, deployed

Postscript - the fifth experiment. After the four nulls we ran the remaining evidence-backed direction: knowledge distillation from the Qwen3-4B teacher (600 stories, five-slot conditioned, style-constrained to simple English; SFT 2 epochs on a Colab T4 in 17 seconds). Uniquely among all five methods, this one *did* move the model - train loss on teacher text is 2.94 (vs 0.67 on self-generated text) and held-out perplexity drifts +4.4% - and the movement was *downward*: 7.57 vs 7.94 (paired $n=45$, $t=-1.55$). The 30M student imitates the teacher’s surface style beyond its capacity and loses native fluency - the “imitation learns style, not content” failure mode (Gudibande et al. 2023). 180k distillation tokens against a 600M-token pretraining prior can bend style but cannot teach structure.

Central empirical finding of the post-training study: at 30M scale with weak, noisy AI feedback, the quality headroom demonstrably *exists at the sample level* but is *not trainable into the default distribution* by any of the low-cost methods above. Inference-time search converts that headroom directly; training methods either fail to move the distribution (in-distribution data) or move it in the wrong direction (off-distribution data beyond the student’s capacity). The default distribution sits at a local optimum determined by pretraining - raising the *floor* requires better pretraining (more/cleaner data, more capacity), not cheap post-training.

9.14 Free vs conditioned generation

Does conditioning on the five slots merely constrain the model, or does it improve the output? We compare 20 *free* fables (no slots, only “write a children’s fable”) against 20 *conditioned* fables (all five slots filled), same model (Phase 2 + DPO) and sampling.

Metric	Free	5-slot conditioned
Distinct-1 / Distinct-2 (diversity)	0.264 / 0.709	0.278 / 0.729
Self-BLEU (repetition, lower better)	0.007	0.005
Flesch reading ease	76.4	80.9 (into children band)
Average length (words)	266	267
Completeness	100%	100%
Slot recall (adherence)	n/a	75%

Finding: conditioning is not just a control interface. Free generation drifts toward a narrower set of stock fables (lower diversity, higher self-similarity across the set, slightly harder readability), whereas the five slots *diversify* the output (they force different characters/settings/challenges) and *ground* it to the user’s request. The scaffold improves quality, not only controllability.

9.15 Efficiency

10. Automatic verdict (Phase 2)

Thresholds are heuristics calibrated for this setup (30M params, 12k vocab, TF1). The dashboard produces a per-metric PASS / WARN / FAIL verdict:

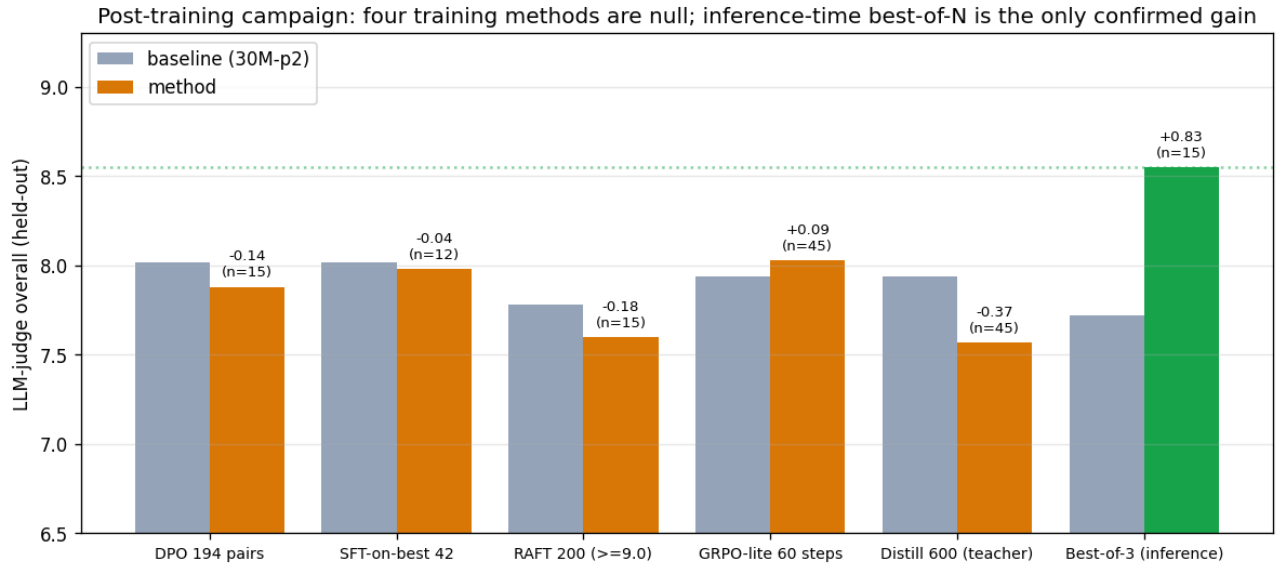


Figure 16: Four training methods, one evaluation protocol, four nulls - and the inference-time search that works. Judge-eval means per experiment pair; n as annotated.

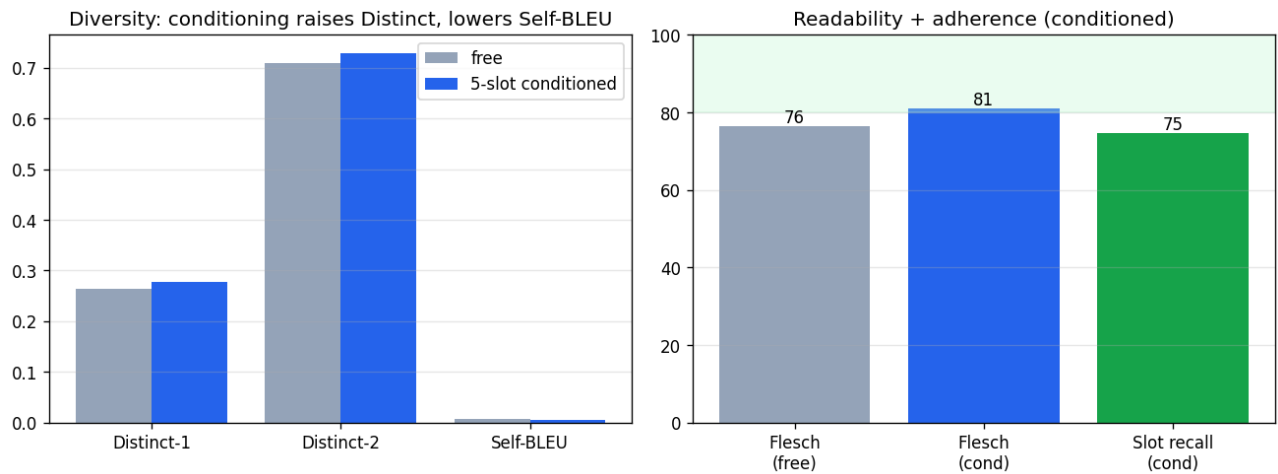


Figure 17: Free vs 5-slot conditioned generation. Conditioning raises cross-set diversity (Distinct up, Self-BLEU down), improves readability into the children band, and grounds the story to the request (75% slot recall) - at no cost to completeness or length.

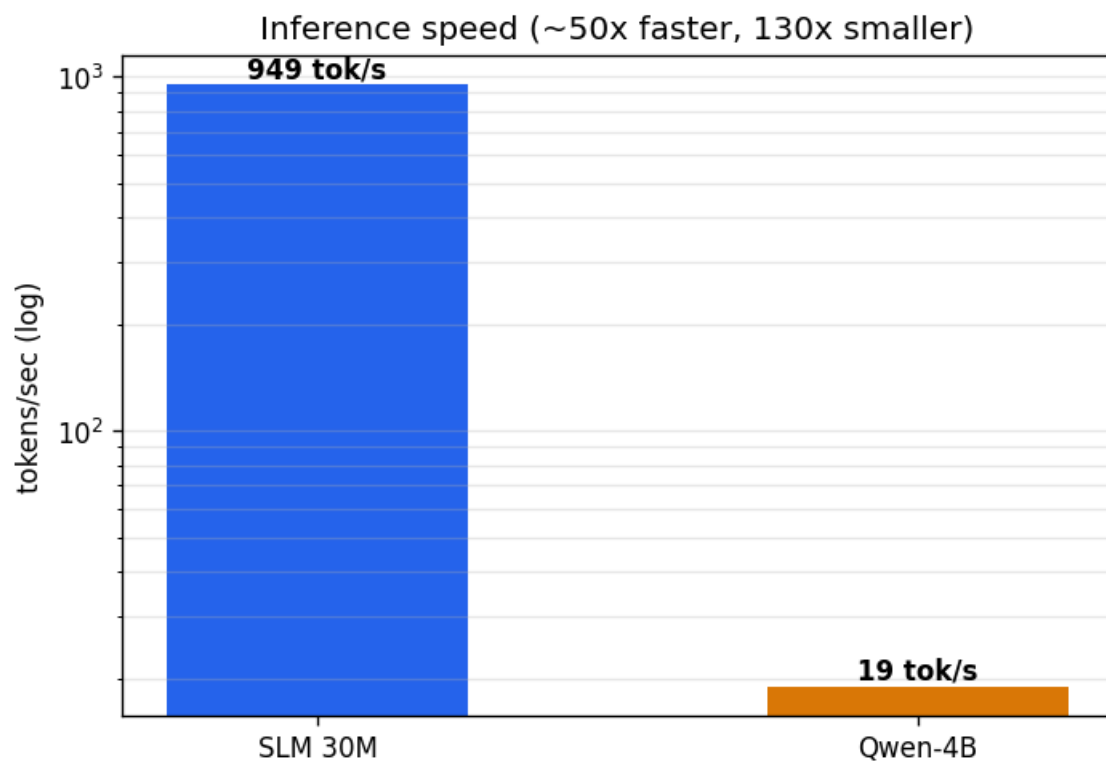


Figure 18: Inference speed. The 30M SLM generates ~949 tokens/s versus ~19 for the 4B reference on the same machine - roughly 50x faster at ~1/130th the parameters.

Metric	Value	Verdict
Final train loss	1.278	PASS
Scaling-law fit R^2	0.959	PASS
Held-out perplexity	3.56 (0.99x floor)	PASS
Distinct-1 gap vs real	8%	PASS
Distinct-2 gap vs real	4%	PASS
Self-BLEU abs gap	0.001	PASS
Flesch reading ease	79.9	WARN (0.1 below band)
Length distribution overlap	43%	WARN
Owl template rate (gen)	23%	PASS

7 PASS / 2 WARN / 0 FAIL.

11. Strengths and limitations

11.1 Strengths

- **Coherent, complete, on-domain output.** After the completeness fix, 30/30 test stories end with a real resolution; generated text matches real fables on diversity (Distinct-2 gap ~4%), repetition (Self-BLEU gap ~0.001), readability (Flesch 79.9 vs 80.0) and the Zipfian vocabulary profile - the model learned the statistical shape of the domain.
- **Efficiency.** ~949 tokens/s, roughly **50x faster** than the 4B reference at **1/130th** the parameters; runs on a laptop with a 39 MB q8 file.
- **Well-generalized language model.** Held-out perplexity 3.56 sits essentially on the theoretical floor $e^{(\text{train loss})}$ - no over-fitting despite the small size.
- **The scaffold improves quality, not only control.** Conditioning on the five slots *raises* cross-set diversity, *improves* readability into the children band and *grounds* the story to the request (Section 9.9) - versus free generation, which collapses toward a narrower set of stock fables.
- **Best-of-N converts headroom into shipped quality.** The +0.8 judge-point gain (7.7 → 8.55) is the study's one confirmed post-training improvement, costs no training, and is exposed in the app as a user control with per-candidate score logging.
- **Negative results are measured, not assumed.** Four alignment methods were tested under one fixed protocol with paired seeds, repeated measurements and an $n=45$ confirmation step; the judge's own noise was quantified (± 0.4 at $n=15$) and used to retract two early false positives - the evaluation methodology is itself a contribution.
- **Reproducible and defensible.** Every number is tied to a run and a published method; the training pipeline, data interventions and evaluation are all scripted.

11.2 Limitations

- **Context ceiling (512 tokens):** the model cannot produce coherent fables beyond ~300-340 words; the length selector has little effect on the SLM.
- **Prompt-adherence ceiling (~70-80%):** a 30M model has weak conditioning; it reads the five slots but still drops one or two (especially abstract Challenge/Outcome) - a quantified size/capability trade-off that none of the tested alignment methods moved.
- **The default distribution resists cheap self-feedback.** DPO, SFT-on-best, RAFT and GRPO-lite (at a ~960-judge-call budget) all fail to shift default-generation quality; only inference-time

selection captures the headroom. Improving the *default* output likely requires an external teacher or a much larger, cleaner reward budget.

- **The evaluation judge is noisy:** repeated evaluations of the same checkpoint differ by up to 0.45 at $n=15$. All headline deltas in this report respect that noise floor; readers should apply the same caution to any small reported difference.
- **Weak length control:** the model has a natural length (~250-280 words) and largely ignores the short/medium/long hint.
- **Template / redemption priors:** the model inherits TF1 biases (friendship/kindness morals; happy endings; a recurring “wise old owl” mediator), which the data intervention reduces but does not eliminate.
- **Local logical slips:** occasional pronoun/reference errors and small non-sequiturs within a story - a capacity limit, not a sampling artifact.
- **Single-judge in-app eval:** the per-generation score is a quick indicator; the headline conclusions use the offline cross-family judge panel with weighted Cohen’s kappa and Kendall’s tau (ADR-0002).

12. Future directions (with feasibility evidence)

Each direction below is paired with concrete evidence from this study that it is likely to pay off, not just a wish-list item.

- **Train to the full Chinchilla budget (~7,900 steps / ~600M tokens).** *Evidence:* at 3,600 steps the loss still lies on the power-law line (R^2 0.96, Fig. 4) and has not plateaued; Kaplan/Chinchilla predict continued gains from more tokens. *Feasibility:* the 400k unique corpus with ≤ 4 -epoch repeat (Muennighoff 2023) supplies the tokens; only more T4 time is needed.
- **Distillation at pretraining scale, not SFT scale.** *Evidence:* the 600-story hard-label distillation moved the model but degraded it (Section 9.13) - the imitation trap. The literature remedy is scale and soft labels: mix teacher data into *pretraining* (millions of tokens, as TF1 itself was Gemma-generated) or distill token-level soft targets (Hinton 2015) rather than fine-tuning on a few hundred hard-label stories. *Feasibility:* T4 SFT takes seconds (measured 17 s for 1,200 samples); the cost is teacher generation time, which scales linearly and parallelizes.
- **Scaled reward-based RL with a cheaper, cleaner reward.** *Evidence:* GRPO-lite was null with $KL \sim 1e-3$ - the policy never moved at a 960-call budget - so the method is untested at scale rather than refuted (Section 9.12). DeepSeek- R1 demonstrates the mechanism works given thousands of steps. *Feasibility:* requires replacing the ~ 15 s/call judge: either a stronger local judge distilled into a fast scorer trained on far more labels (the 30M scorer failed at ~ 500 labels; scaling labels is the testable fix), or rule-based partial rewards (slot recall, completeness) that cost microseconds.
- **Stronger evaluation judge and larger eval sets by default.** *Evidence:* the measured ± 0.4 noise at $n=15$ (Section 9.11) manufactured two false positives; $n=45$ paired evals resolved both. *Feasibility:* the 36 GB machine runs a 14B judge comfortably; $n=45$ -100 evals are an hour of compute.
- **Targeted data debiasing beyond the owl template.** *Evidence:* the single “wise old owl” cap already moved the generation rate 90% $\rightarrow$ 23% (Fig. 10), proving the intervention mechanism works; the same recipe can address the happy-ending/redemption prior and other stock phrases.
- **Extended context / explicit length control.** *Evidence:* the 512-token ceiling is the direct cause of both the length-control weakness and rare truncation (Section 9.6). *Feasibility:* re-training at 1024 tokens or adding length-bucket control tokens is a known, bounded change.

13. Conclusion

A 30M-parameter model trained from scratch, guided by scaling-law reasoning and a targeted data intervention, reaches $\sim 7.9/10$ fable quality (final protocol, $n=45$; from 2.5 at the under-trained baseline) with held-out perplexity 3.56, generating complete, on-domain stories at $\sim 50\times$ the speed and $1/130$ th the size of a 4B LLM. A systematic post-training campaign then delivers the study's sharpest finding: the model's quality headroom is real at the sample level (best-of-3 reaches 8.55, near the 4B reference at 9.75) but **no low-cost post-training method moves the default distribution upward** - DPO, SFT-on-best, RAFT and budget-limited GRPO are null, and teacher distillation at SFT scale moves it *down* (the imitation trap) - while inference-time best-of-N captures the gain directly and ships in the app. The result supports the thesis with an honest boundary: **for a well-scoped task, a tiny model can rival a large one on the axes that matter, at a fraction of the cost** - provided the remaining variance is managed at inference time, and with the size ceiling documented where it shows (length control, residual adherence gap, alignment resistance).

14. Reproducibility

All model code lives under `trieulh/` (isolated from the shared web app):

- Data + training: `trieulh/scripts/prepare_tf1_pretrain.py`, `train_tokenizer.py`, `tf1_pretrain/`; notebook `trieulh/notebooks/pretrain_slm_30m_dashboard.ipynb`.
- Alignment campaign: `trieulh/scripts/gen_preference_pairs.py`, `dpo_train_local.py`, `headroom_probe.py` (best-of-N), `raft_harvest.py` + `raft_gen_corpus.py` + `sft_best_local.py` (RAFT), `rm_train.py` + `rm_train_pairwise.py` (reward-model gate), `grpo_train.py` (GRPO-lite), `raft_judge_eval.py` / `grpo_judge_eval.py` / `big_judge_eval.py` (the shared evaluation protocol).
- Evaluation: `trieulh/scripts/eval_slm.py`; app metrics `app/metrics.py`, `app/perplexity.py`.
- Experiment log (source of this report): `trieulh/docs/experiments/2026-07-08-slm-training-log.md`.
- Artifacts (Google Drive): checkpoints, HF models (30M, 30M-p2, 30M-dpo), GGUF exports, `analysis_*.json`, `loss_log_*.json`, and the dashboard figures.

Model download. The final aligned model (Phase 2 + DPO) is packaged as a zip (GGUF q8 + Ollama Modelfile + HF checkpoint): <https://drive.google.com/file/d/1tY6dPodSqHunYIYEZDdMOEZ1dJyg2HuD/view?usp=drivesdk> To run it: `unzip`, then `ollama create slm-30m-dpo -f Modelfile-30M-dpo`.

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